



RESEARCH ARTICLE

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Key Points:

- We build a new step in the chemistry-climate model hierarchy that combines full-complexity chemistry with simplified climate dynamics
- An idealized model successfully reproduces key aspects of the global OH budget as simulated in a fully coupled chemistry-climate model
- The OH response to climate warming is governed by two competing effects: moistening and temperature-sensitive biogenic emissions

Supporting Information:

Supporting Information may be found in the online version of this article.

Correspondence to:

Q. Zhu,
qindan.zhu@cfa.harvard.edu

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Author Contributions:

Conceptualization: Qindan Zhu, Nicole Neumann, Arlene M. Fiore, Robert Pincus, Clare E. Singer, Brian Medeiros

Data curation: Qindan Zhu, Nicole Neumann

Formal analysis: Qindan Zhu, Nicole Neumann, Jian Guan, George Milly, Clare E. Singer

Funding acquisition: Arlene M. Fiore, Robert Pincus

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Uncertain Natural Emissions Dampen the Increase in Tropospheric Hydroxyl Radical (OH) With Idealized Surface Warming

Qindan Zhu^{1,2} , Nicole Neumann^{3,4} , Arlene M. Fiore^{1,3} , Robert Pincus³ , Jian Guan¹ , George Milly³ , Clare E. Singer³ , Brian Medeiros⁵ , and Paolo Giani¹

¹Department of Earth, Atmospheric and Planetary Sciences, Massachusetts Institute of Technology, Cambridge, MA, USA,

²Now at Atomic and Molecular Physics Division, Center for Astrophysics, Harvard & Smithsonian, Cambridge, Cambridge, MA, USA, ³Lamont-Doherty Earth Observatory (LDEO), Columbia Climate School, Palisades, NY, USA, ⁴Now at Department of Oceanography, Texas A&M University, College Station, TX, USA, ⁵Climate and Global Dynamics Laboratory, National Center for Atmospheric Research, Boulder, CO, USA

Abstract The hydroxyl radical (OH) defines the oxidative capacity of the atmosphere and determines the lifetime of reactive greenhouse gases, including methane. The response of OH to climate warming is influenced by uncertain and compensating processes involving meteorological factors and temperature-sensitive natural emissions, including soil NO_x (SNO_x) and biogenic volatile organic compounds (BVOC) emissions. However, separating individual processes that control the OH response to warming is challenging given the high dimensionality of both climate dynamics and emissions in fully coupled chemistry-climate models. Here, we create an idealized chemistry-climate model, Aqua-chem, by prescribing annual mean emissions and zonally symmetric sea surface temperatures. We show that the net OH response to an idealized 2 K surface warming in Aqua-chem depends on competing effects of moistening (a robust response to warming) and temperature-sensitive BVOC emissions (a highly uncertain response). The 2 K surface warming increases water vapor, resulting in an increase in tropospheric OH through primary OH production (ozone photolysis followed by reaction of O¹D with H₂O). Temperature-sensitive SNO_x emissions further enhance OH via the NO + HO₂ reaction, but this additional increase is outweighed by the increase in temperature-sensitive BVOC emissions. Amplified OH losses, through reactions with BVOCs and their oxidation byproducts, strongly dampen the increase due to atmospheric moistening with rising surface temperature. Our study underscores the importance of accurately quantifying the temperature sensitivity of natural emissions in order to constrain the OH response to climate warming.

Plain Language Summary Methane is mainly removed from the atmosphere through the reaction with the hydroxyl radical (OH). As the climate warms, changes in OH directly impact how quickly methane is broken down, affecting the methane budget. However, OH is controlled by a complex, non-linear system that involves both climate dynamics and atmospheric chemistry, making it difficult to predict how OH will respond to warming. In this project, we developed an idealized chemistry-climate model, Aqua-chem, which simplifies the climate dynamics while preserving the full complexity of atmospheric chemistry. This model allows us to quickly and effectively assess how OH responds to changes in surface temperature. Our results show that rising surface temperature increases water vapor, a key limiting factor for OH production in much of the atmosphere. However, warming also leads to higher biogenic emissions of reactive carbon compounds, which increase the OH loss frequency. Ultimately, the sign and magnitude of OH changes under climate warming are shaped by these two competing effects, the increase in OH due to moistening and the decrease in OH due to enhanced biogenic emissions.

1. Introduction

Methane (CH₄) is second only to carbon dioxide (CO₂) as anthropogenic greenhouse gas (GHG), with its atmospheric concentration increasing by ~160% since pre-industrial times and contributing $+0.54 \pm 0.11 \text{ W m}^{-2}$ to effective radiative forcing (Szopa et al., 2021). Climate warming induced by CH₄ and other GHGs, in turn, affects future CH₄ projections (Dean et al., 2018; Holmes et al., 2013). Climate feedback on CH₄ occurs through weather-sensitive natural CH₄ emissions such as wetlands (e.g., Bridgman et al., 2013; Gedney et al., 2004; Shindell et al., 2004), and its primary chemical loss through reaction with the hydroxyl radical (OH) (Lelieveld &

Investigation: Qindan Zhu, Nicole Neumann, George Milly, Clare E. Singer, Brian Medeiros, Paolo Giani
Methodology: Qindan Zhu, Nicole Neumann, Arlene M. Fiore, Robert Pincus, Jian Guan, George Milly, Clare E. Singer, Brian Medeiros, Paolo Giani
Project administration: Arlene M. Fiore, Robert Pincus
Resources: Qindan Zhu
Software: Qindan Zhu, Nicole Neumann, Jian Guan, George Milly, Brian Medeiros
Supervision: Qindan Zhu, Arlene M. Fiore, Robert Pincus
Validation: Qindan Zhu, Nicole Neumann, Jian Guan, Clare E. Singer
Visualization: Qindan Zhu
Writing – original draft: Qindan Zhu
Writing – review & editing: Qindan Zhu, Nicole Neumann, Arlene M. Fiore, Robert Pincus, Jian Guan, George Milly, Clare E. Singer, Brian Medeiros, Paolo Giani

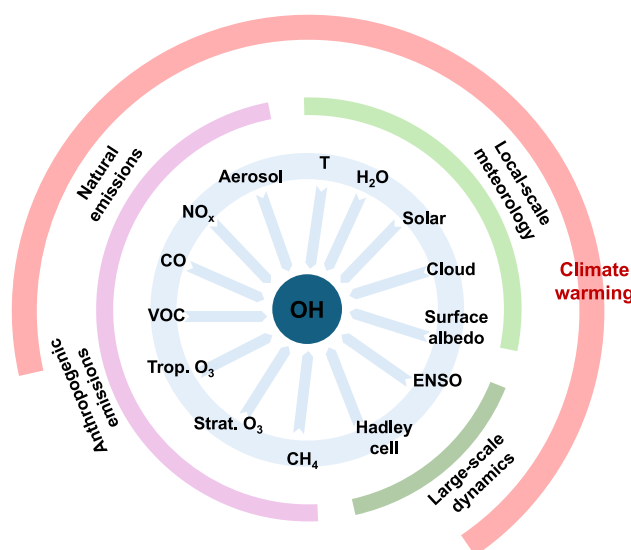


Figure 1. The OH response to climate warming occurs through changes in meteorology and temperature-sensitive emissions. As reported in previous studies (Dentener et al., 2003; Ehhalt et al., 2001; Fiore et al., 2006; He et al., 2021; Hess & Mahowald, 2009; Holmes et al., 2013; Lelieveld & Crutzen, 1994; Nicely et al., 2018, 2020; Prinn et al., 2001; Spivakovsky et al., 2000; Voulgarakis et al., 2010), OH is influenced by meteorological parameters (such as temperature, water vapor, solar radiation, surface albedo, ENSO, and Hadley cell dynamics) as well as by chemical factors (e.g., aerosols, NO_x , CO, VOCs, tropospheric and stratospheric O_3 , CH_4), with the latter being determined by both anthropogenic and natural emissions. Climate warming impacts meteorology and emissions, leading to changes in OH chemistry.

Crutzen, 1994; Spivakovsky et al., 2000). The CH_4 chemical loss depends on OH concentration and the reaction rate between OH and CH_4 ($k(\text{CH}_4 + \text{OH})$). While $k(\text{CH}_4 + \text{OH})$ is highly temperature-dependent and well-understood, the response of OH to climate warming remains uncertain.

OH chemistry is highly non-linear and is driven by emissions and meteorological factors (Dentener et al., 2003; Ehhalt et al., 2001; Fiore et al., 2006; He et al., 2021; Hess & Mahowald, 2009; Holmes et al., 2013; Lelieveld & Crutzen, 1994; Nicely et al., 2018, 2020; Prinn et al., 2001; Spivakovsky et al., 2000; Voulgarakis et al., 2010). For instance, emissions of carbon monoxide (CO), nitrogen oxides (NO_x), and volatile organic compounds (VOCs) contribute to OH production through HO_x cycling and O_3 photolysis followed by reaction with H_2O , while simultaneously serving as OH sinks through direct oxidation. OH also depends on local meteorological parameters such as temperature, circulation, water vapor, stratospheric ozone, and clouds. In addition, large-scale dynamical features, including the El Niño-Southern Oscillation (ENSO) and Hadley cell expansion, play important roles in regulating OH variability (Nicely et al., 2018; Turner et al., 2018).

Climate warming impacts OH chemistry by altering meteorological factors; it also influences temperature-sensitive natural emissions (Figure 1), including wetland CH_4 emissions, biogenic VOC (BVOC) and soil NO_x (SNO_x) emissions. BVOCs encompass a broad range of VOCs emitted naturally from vegetation, such as isoprene and monoterpenes, which are more reactive than many anthropogenic VOCs. Soils leak NO to the atmosphere during nitrification and denitrification (Pilegaard, 2013). The magnitude and sign of the response of wetland CH_4 emissions to climate warming is uncertain and varies across models (Dean et al., 2018; Kleinen et al., 2021; Melton et al., 2013). BVOC and SNO_x emissions increase with climate warming, as parameterized in the models (Guenther et al., 2012; Hudman et al., 2012). However, given their sensitivities to various other ecological and environmental factors, such as microbial activity, soil moisture, soil type, and elevated CO_2 concentrations, the net influence of climate warming on both BVOC and SNO_x emissions remains uncertain, as is their impact on the global OH budget (Lee et al., 2024; Pilegaard, 2013).

In fully coupled chemistry-climate models, disentangling the role of specific processes in controlling the response of OH to climate warming is challenging due to the high dimensionality and associated uncertainties in emission and meteorological factors. Here, we aim to build an idealized chemistry-climate model that includes the first-order controls on emissions and climate dynamics to provide a fundamental yet still quantitative investigation of the impacts of specific processes and their uncertainties on the OH response to planetary surface warming. This

model framework draws on the concept of a “climate model hierarchy” used extensively in the climate community (Held, 2005). For example, the CESM2 aquaplanet framework is developed by replacing the land and ocean models with prescribed, zonally-symmetric sea surface temperature fields with an equator-to-pole temperature gradient, which greatly simplifies the resulting atmospheric dynamics (Medeiros, 2020; Medeiros et al., 2016). It has proven effective for building process-level understanding of how physical temperature sensitivity depends on specific processes, such as cumulus parameterizations (Miura et al., 2005), cloud feedbacks (Medeiros & Stevens, 2011; Medeiros et al., 2008), as well as for probing controls on tropical cyclone activity (Zhang et al., 2021) and interactions between resolved and parameterized scales (Williamson, 2008).

In this study, we conduct simulations with two models: a full complexity chemistry–climate model (CAM-chem; Emmons et al., 2020) and a newly developed idealized, water-covered chemistry–climate model (Aqua-chem). Aqua-chem is configured by extending the CESM2 aquaplanet configuration to include atmospheric chemistry, for which we prescribe emissions and deposition velocities, including those based on land. Using the same chemistry configuration as CAM-chem, we examine how the simplified climate dynamics affect model representation of OH chemistry simulated by CAM-chem (Section 2). Additionally, we analyze the OH budget by examining individual OH production and loss terms and diagnose the specific drivers of the OH response to an idealized +2 K surface warming. We investigate the roles of temperature-sensitive natural emissions, including BVOC and SNO_x emissions, in regulating OH changes due to climate warming (Section 3).

2. A New Idealized Model to Study Chemistry–Climate Interactions

We choose the Community Earth System Model (CESM) to build a new step in a chemistry–climate modeling hierarchy. We select CESM because it offers a range of model configurations including an idealized aquaplanet configuration (Medeiros et al., 2016) and some of the most complex chemical mechanisms (Emmons et al., 2020; Schwantes et al., 2020) used in fully coupled chemistry–climate models (Danabasoglu et al., 2020; Gettelman et al., 2019). Furthermore, ongoing development efforts are adding modularity that enables the implementation of additional chemical mechanisms (Fritz et al., 2022; Lin et al., 2024).

2.1. CAM-chem

The Community Atmosphere Model with Chemistry (CAM-chem) refers to the implementation of atmospheric chemistry in CESM2-CAM6. Configured with the MOZART-T1 chemical mechanism (Emmons et al., 2020) and the volatility basis set parameterization for the formation of secondary organic aerosols (SOA) (Tilmes et al., 2019), CAM-chem is used to simulate global tropospheric and stratospheric atmospheric composition (Lamarque et al., 2012). In CAM-chem, we prescribe the sea surface temperatures (SSTs) and sea-ice distribution. The land component is fully active and provides the chemistry with deposition velocities and biogenic non-methane volatile organic compounds (NMVOCs) emissions. Biomass burning and anthropogenic emissions, including soil NO_x emissions categorized as agricultural sources, are obtained from the Coupled Model Inter-comparison Project Phase 6 (CMIP6) (Feng et al., 2020; Hoesly et al., 2018; Van Marle et al., 2017). The biogenic NMVOC emissions are computed online by the Model of Emissions of Gases and Aerosols from Nature (MEGAN) v2.1 scheme (Guenther et al., 2012). Lightning NO is produced based on the parameterization described in Price et al. (1997) and Lamarque et al. (2012).

2.2. Aqua-chem

Here, we extend the CAM6 aquaplanet configuration to include atmospheric chemistry and associated emissions (Table 1). Similar to the original aquaplanet design, Aqua-chem is driven by idealized zonal-mean, annual-mean, and hemispherically symmetric SSTs (Figure S1 in Supporting Information S1), reducing the complexity and variability of the weather, while still retaining the key features of the atmospheric large-scale circulation such as the Hadley Cell and upper tropospheric jets. However, due to the absence of seasonal cycle and sea ice in the polar regions, Aqua-chem cannot accurately represent dynamics at the poles. Consequently, our study domain is confined to latitudes between 60°N and 60°S, where OH is predominantly located.

Although Aqua-chem is entirely water-covered, it is configured as Earth-like for atmospheric chemistry in that we incorporate the predominantly land-based anthropogenic, biogenic, and biomass-burning emissions from CAM-chem. While biogenic NMVOC emissions are calculated online in CAM-chem, in Aqua-chem we simply prescribe the biogenic emissions simulated by CAM-chem similar to the treatment of anthropogenic emissions and as

Table 1
Model Configurations Used in This Study

	CAM-chem	Aqua-chem
Chemistry	MOZART-T1 mechanism	
CH ₄	Lower boundary condition (1,773 ppb)	
Planetary surface	Topography	Water-covered
SSTs	Prescribed	Prescribed annual-mean zonal-mean
Emissions	Prescribed/online calculation	Annual-mean emissions from CAM-chem

Note. SSTs refer to sea surface temperatures. The parameterizations of physical processes, including planetary boundary layer (PBL) mixing, are identical between Aqua-chem and CAM-chem. The cloud-aerosol online interaction is disabled in Aqua-chem.

done in some early atmospheric chemistry models (e.g., Emmons et al. (2010)). We also prescribe dry deposition velocities. Given the absence of seasonality in Aqua-chem, we prescribe annual mean emissions. Lightning NO_x emissions differ between CAM-chem and Aqua-chem: the absence of land leads to lightning NO emissions that are an order of magnitude lower in Aqua-chem than those in CAM-chem (0.11 vs. 3.2 Tg N/year, Figure S2 and Section S1 in Supporting Information S1).

Due to the simplification in climate dynamics related to OH chemistry, Aqua-chem allows for a much shorter integration time compared to the reference model, CAM-chem. While CAM-chem requires a full year of simulation to capture annual OH chemistry, the Aqua-chem model can achieve this in just 1 month due to its zonal symmetry in climate dynamics and absence of seasonal variability. As a result, the computational expense of Aqua-chem is reduced by an order of magnitude compared to CAM-chem, enabling efficient model simulations to explore OH chemistry sensitivity to climate warming and to isolate the effects of natural emissions on the OH budget.

All simulations conducted in this study use the same horizontal resolution of $1.25^\circ \times 0.95^\circ$ and 32 vertical layers. The CAM-chem experiments start with a 1-year spin-up period that is discarded, followed by 30 years. We select a 30-year period, consistent with NOAA NCEI's recommended standard for defining climate normals (<https://www.ncei.noaa.gov/access/us-climate-normals/>). The Aqua-chem experiments are run for 3.5 years, including a 1-year spin-up period. The SSTs are prescribed in all simulations. The CAM-chem experiments apply a repeating climatological monthly-mean SST centered on the year 2000, while Aqua-chem applies an idealized SST pattern that depends only on latitude (Figure S1 in Supporting Information S1). Since the forcing is a fixed, repeating signal, one can consider the CAM-chem experiments as 30 independent realizations of the annual cycle. Similarly, since the Aqua-chem setup is symmetric with constant forcing, the 30 months used for analysis provide analogous sampling of internal climate variability to the much longer CAM-chem experiments.

Distributions of longer-lived species (those with lifetimes exceeding 1 month, such as CO and C₂H₆) rapidly reach a steady spatial distribution in Aqua-chem through the combined effects of emissions and transport under the simplified and much less variable dynamics of Aqua-chem (Figure S4 in Supporting Information S1). Moreover, because the full Aqua-chem simulation spans 3.5 years, we discard the first year as spin-up, ensuring sufficient time for these chemical species to equilibrate and adjust to the imposed perturbations.

Below, we show that despite simplified climate dynamics and annually prescribed emissions, Aqua-chem successfully reproduces key terms in the OH budget (Section 2.3) and the response of the OH budget to climate warming (Section 2.4) simulated in CAM-chem.

2.3. Evaluation of the Present-Day OH Budget in Aqua-chem

We conduct present-day simulations in both CAM-chem and Aqua-chem. The CAM-chem simulation is configured with SST and emissions fixed at climatological year 2000 levels, based on the 1995–2005 observed decadal average. The Aqua-chem simulation is configured using the annual-mean emissions from CAM-chem and zonal average SSTs at the year 2000 level.

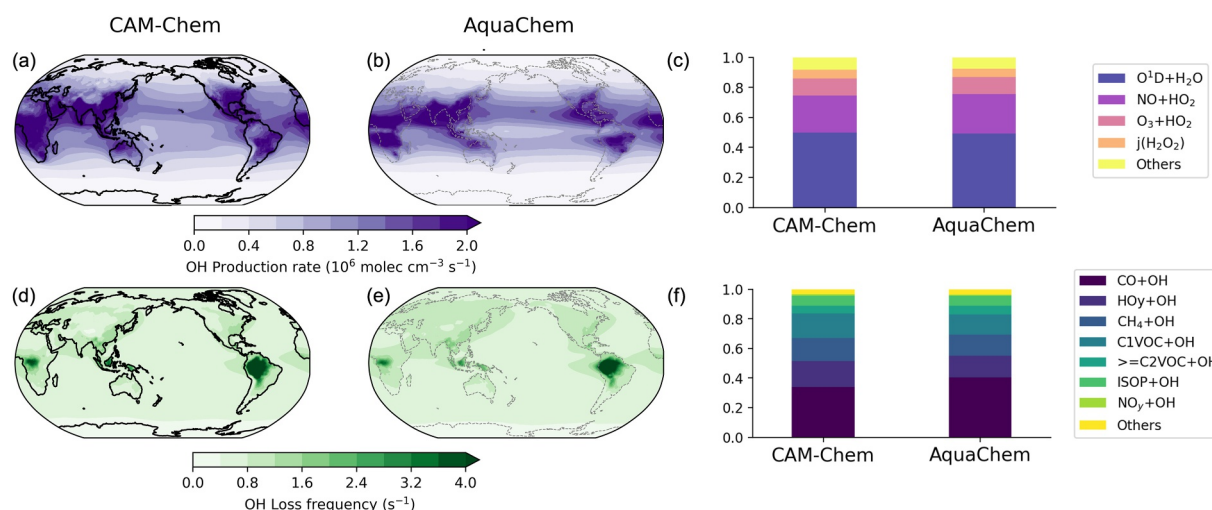


Figure 2. The idealized Aqua-chem model captures the salient spatial patterns and partitioning among OH production rate and loss frequency in the full complexity CAM-chem model. Spatial distributions of OH production rates (a–b) and OH loss frequencies (d–e) simulated by CAM-chem and Aqua-chem. Panels (c and f) compare the contributions of individual OH production and loss pathways to the total OH production rate and OH loss frequency, respectively. The production pathways include reactions $\text{O}^1\text{D} + \text{H}_2\text{O}$, photolysis of H_2O_2 , $\text{NO} + \text{HO}_2$, $\text{O}_3 + \text{HO}_2$ and others (e.g., ozonolysis of alkene). The loss pathways include OH reactions with single-carbon non-methane volatile organic compound, VOC with >2 C atoms, CH_4 , CO , HO_y (H_2 , O_3 , H_2O_2 , radical–radical reactions), biogenic VOCs (BVOC), NO_y (NO_2 , N-reaction products), and others (e.g., aerosol uptake). Gray dashed continental outlines in Aqua-chem indicate that land-based drivers (e.g., emissions, deposition) are included for chemistry but not for climate.

Due to its short lifetime of $<1\text{s}$, OH is at a steady state, with a balance between its production and loss (Equation 1). In this analysis, in addition to air-mass-weighted OH, we calculate the air-mass-weighted OH production rate and air-mass-weighted OH loss frequency to diagnose the drivers of OH chemistry.

$$\text{Rate}_{\text{prod}} = \text{Rate}_{\text{loss}} = \text{OH} \times \text{Freq}_{\text{loss}} \quad (1)$$

Even though Aqua-chem produces a global mean OH concentration that is 30% lower than that in CAM-chem (Figures S11a and S11b in Supporting Information S1), it successfully reproduces the OH production rates simulated by CAM-chem. The spatial distributions of the annual air-mass-weighted OH production rate in CAM-chem and Aqua-chem correlate strongly (Figures 2a and 2b), correlation coefficient (R^2) of 0.92, Figure S8 in Supporting Information S1). The global OH production rate is $1.1 \times 10^6 \text{ molec cm}^{-3} \text{ s}^{-1}$ in CAM-chem and $1.0 \times 10^6 \text{ molec cm}^{-3} \text{ s}^{-1}$ in Aqua-chem. Shown in Figure 2c, Aqua-chem reproduces the relative contributions of individual OH production pathways. In both CAM-chem and Aqua-chem, four dominant pathways account for 92% of the total OH production rate. The primary OH production pathway, O_3 photolysis followed by its reaction with H_2O , contributes 50% of the total OH production rate. The $\text{NO} + \text{HO}_2$ reaction accounts for 25%, while the $\text{O}_3 + \text{HO}_2$ reaction and H_2O_2 photolysis contribute 11% and 6%, respectively.

Similarly, Aqua-chem effectively represents OH loss frequency. As shown in Figures 2d and 2e, CAM-chem and Aqua-chem exhibit a consistent spatial distribution of the annual air-mass-weighted OH loss frequency, with an R^2 of 0.90. The global OH loss frequency in Aqua-chem is 0.82 s^{-1} , closely aligning with the CAM-chem simulation of 0.80 s^{-1} . In CAM-chem the OH reaction with CO accounts for 34% of the total OH loss frequency, while OH reactions with HO_y (H_2 , O_3 , H_2O_2 , radical–radical reactions), CH_4 , and single-carbon NMVOCs each contribute 14%. Compared to CAM-chem, Aqua-chem simulates a 6% higher contribution of the $\text{CO} + \text{OH}$ reaction, but it maintains the relative ranking of OH loss pathways (Figure 2f).

2.4. Evaluation of the OH Budget Response to Idealized Surface Warming in Aqua-chem

We conduct idealized warming experiments in both CAM-chem and Aqua-chem, configured with a uniform increase in SST of 2 K globally. Since the BVOC emissions in CAM-chem are calculated online using MEGAN, this idealized warming simulation in CAM-chem (“CAM-chem, Met + BVOC”) accounts for the response of the OH budget to both meteorology and temperature-dependent BVOC emissions. Figure S3a in Supporting

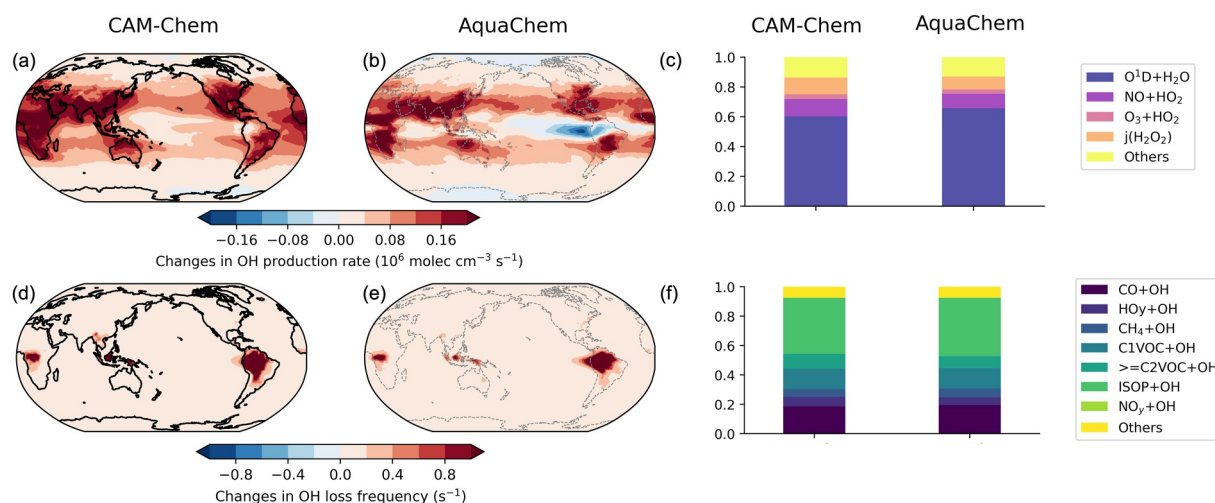


Figure 3. Aqua-chem reproduces the response of OH budget terms to a +2 K idealized surface warming simulated in the full complexity CAM-chem model. Changes in OH production rates (a–b) and OH loss frequencies (d–e) due to a 2 K idealized SST warming, as simulated by CAM-chem and Aqua-chem with temperature-sensitive BVOC emissions and constant SNO_x emissions (“Met + BVOC”). Panels (c and f) compare the contributions of individual OH production and loss pathways, respectively. The individual OH production and loss pathways are labeled as in Figure 2.

Information S1 shows that the global annual isoprene emission simulated by MEGAN (Guenther et al., 2012) increases by 37% in response to the rise in the land surface temperature following the 2 K increase in SST (346 vs. 476 Tg C/year). To validate Aqua-chem's ability to represent these changes, although the OH response is underestimated by 47% (Figure S11c and S11d in Supporting Information S1), we compare the OH budget response in CAM-chem against the Aqua-chem simulation with the enhanced BVOC emissions (“Met + BVOC”).

Figures 3a–3c shows the changes in OH production rate under a 2 K idealized surface warming compared to present day. The spatial distribution of the changes in OH production rate simulated by CAM-chem and Aqua-chem yields an R^2 of 0.74. The global OH production rate increases by $9.8 \times 10^4 \text{ molec cm}^{-3} \text{ s}^{-1}$ in CAM-chem, while Aqua-chem underestimates this increase at $7.2 \times 10^4 \text{ molec cm}^{-3} \text{ s}^{-1}$. The largest biases appear over the tropical Pacific and the Amazon rainforest, which is likely due to Aqua-chem's underestimation of lightning NO emissions in the absence of land (Figure S2 in Supporting Information S1) or differences in the meteorology arising from the simplified treatment of atmospheric dynamics. However, Aqua-chem accurately reproduces the relative contributions of individual OH production pathways to the changes in the OH production rate. In both models, the primary OH production pathway remains the dominant contributor to increased OH production rate, accounting for 60% of the increase in production rate in CAM-chem and 66% in Aqua-chem.

Figures 3d–3f illustrates changes in the OH loss frequency due to the same warming scenario. Under the 2 K idealized warming, the increase in global OH loss frequency is 0.11 s^{-1} in CAM-chem, primarily driven by BVOC-rich regions. Aqua-chem simulates a global increase in the OH loss frequency of 0.10 s^{-1} , with the pattern of change correlated with CAM-chem ($R^2 = 0.84$). Notably, the idealized model captures the CAM-chem attribution of the changes in OH loss frequency to individual OH loss pathways. The largest contribution is 38% from OH reactions with BVOC species, including isoprene and monoterpene.

3. Diagnosis of the Individual Processes Controlling the OH Response to Surface Warming

Here we use Aqua-chem to diagnose the OH response to idealized surface warming via individual processes, including meteorology, SNO_x emissions and BVOC emissions (Figures 4 and 5). In addition to the idealized warming simulation in Aqua-chem with the temperature-sensitive BVOC emissions response (“Met + BVOC”), we conduct another Aqua-chem simulation in which we hold the BVOC emissions constant, so the idealized warming only influences meteorology (“Met only”). Even though CAM-chem is configured with constant soil NO_x emissions, a recently developed soil NO_x scheme, the Berkeley Dalhousie Iowa Soil NO Parameterization

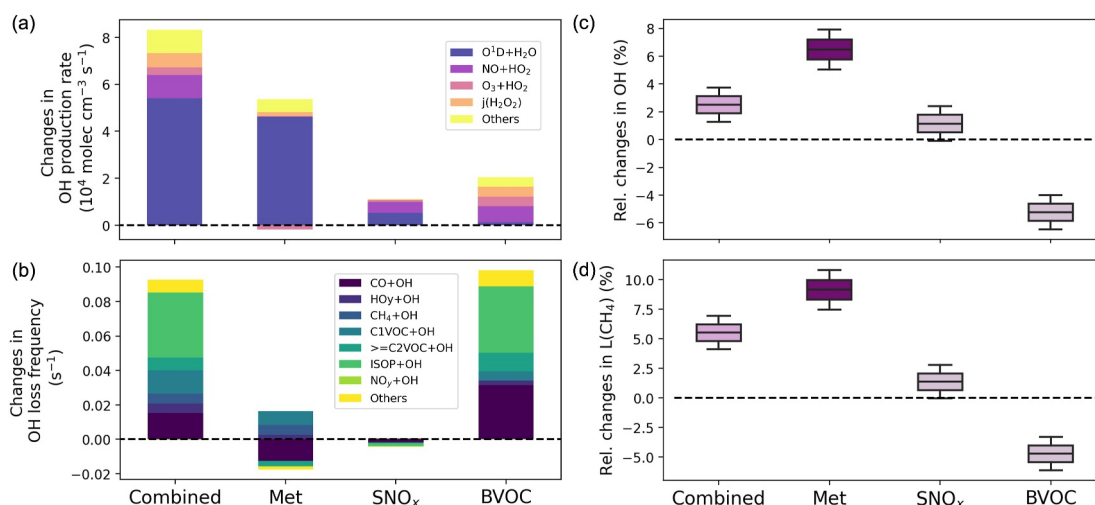


Figure 4. Both meteorology and temperature-sensitive natural emissions influence OH chemistry by modifying the response of individual OH production and loss pathways to idealized warming. Absolute changes in airmass-weighted OH production rates (a), OH loss frequency (b); and relative changes in OH (c) and CH_4 chemical loss ($L(\text{CH}_4)$), (d) in response to a 2 K surface warming in Aqua-chem under the “Combined,” “Met,” “SNO_x” and “BVOC” cases. Purple shadings in (c) and (d) indicate the level of scientific understanding, with the darkest shade representing the highest level of confidence. The individual OH production and loss pathways are defined and labeled as in Figure 2.

(BDISNP) (Hudman et al., 2012; Sha et al., 2021), suggests that soil NO_x emissions increase by 23% under a 2 K idealized warming scenario (Figure S3b in Supporting Information S1, 4.9 vs. 6.0 Tg N/year). Therefore, we explore the impact of temperature-sensitive SNO_x emissions with two additional idealized warming experiments in Aqua-chem (“Met + SNO_x,” “Met + SNO_x + BVOC”).

We attribute the OH budget changes to individual processes by differencing pairs of Aqua-chem simulations driven by base vs +2 K SSTs. “Combined” shows the overall response of the OH and OH budget terms through meteorology and temperature-sensitive natural emissions to the idealized warming scenario (“Met + SNO_x + BVOC”). “Met” shows the response solely from the physical climate (“Met only”). “SNO_x” contrasts the “Met + SNO_x” and “Met only” scenarios to isolate the role of soil NO_x responding to higher surface temperature.

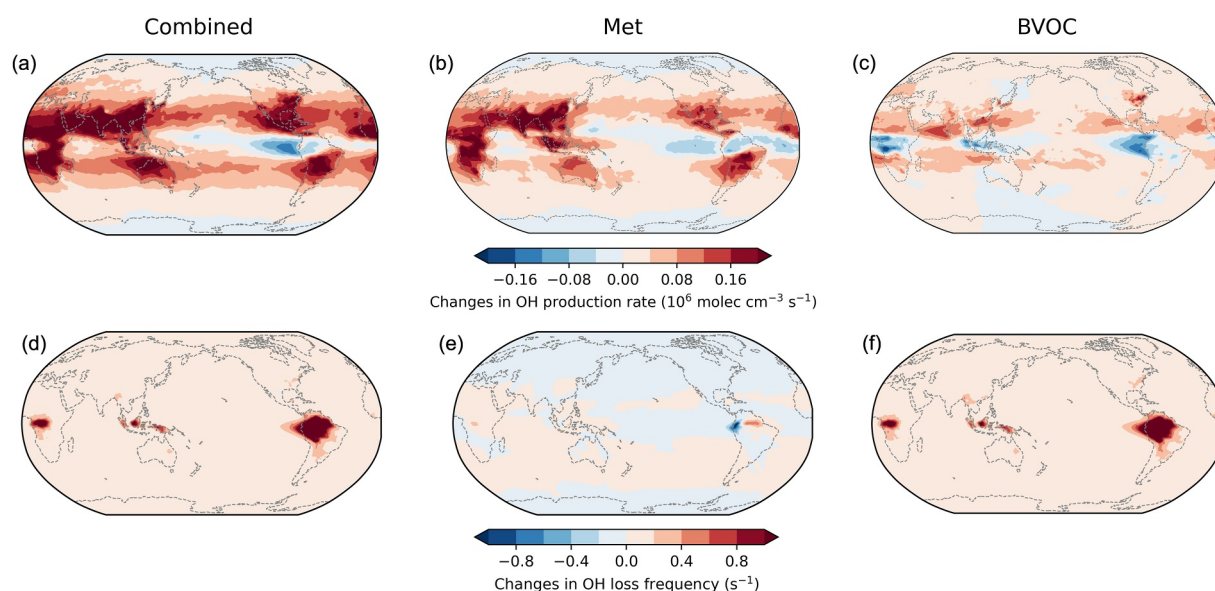


Figure 5. Meteorology and temperature-sensitive BVOC emissions drive the increase in OH production rate and OH loss frequency under an idealized +2 K SST warming, respectively. Changes in OH production rate (a–c) and OH loss frequency (d–f) under the “Combined,” “Met,” and “BVOC” cases, as described in Figure 4.

By differencing the “Met + BVOC” and “Met only” scenarios we examine the effects of temperature-sensitive BVOC emissions on OH chemistry.

3.1. Warming and Moistening (“Met Only”)

OH increases with idealized surface temperature primarily because warming increases water vapor (H_2O). Although Aqua-chem does not reproduce the enhanced surface warming over land (Figure S12 in Supporting Information S1), it yields a consistent increase in H_2O relative to CAM-chem. The global air-mass-weighted tropospheric OH increases by 6.5% in the “Met-only” experiment (Figure 4c). This increase in OH is primarily driven by the increase in the OH production rate (Figure 5b). On average, the global OH production rate increases by $8.4 \times 10^4 \text{ molec cm}^{-3} \text{ s}^{-1}$. The primary OH production pathway accounts for 89% of the total enhancement (Figure 4a). This enhancement reflects a +14–17% increase in H_2O with a 2 K temperature increase, consistent with the Clausius–Clapeyron–inferred 7%/K increase (Figure S5 in Supporting Information S1). The methane chemical loss ($L(\text{CH}_4)$) increases more, by 9.1% due to the combined effects of enhanced OH and the temperature-sensitive rate constant $k(\text{CH}_4 + \text{OH})$ (Figure 4d).

In contrast to the OH production term, the OH loss frequency stays roughly constant under 2 K warming (Figure 5e), due to the offsetting effects among different OH sink pathways (Figure 4b). Under a 2 K warming, even though the OH loss frequency increases through reactions with CH_4 , HO_y , and single-carbon NMVOCs by 0.016 s^{-1} , it is counterbalanced by a reduction in the OH loss frequency of 0.015 s^{-1} through OH reactions with CO and VOCs with ≥ 2 C atoms. The largest reduction in the OH loss frequency is attributed to the OH reaction with CO. Under the idealized warming with constant natural emissions, the increase in the CO sink due to enhanced OH outweighs the increase in CO production from the $\text{CH}_4 + \text{OH}$ reaction, leading to a net reduction of 2 ppb in CO (Figure S6a in Supporting Information S1). Consequently, the OH loss frequency through the $\text{CO} + \text{OH}$ reaction decreases by 0.013 s^{-1} .

3.2. Temperature-Sensitive Natural Emissions

Temperature-sensitive SNO_x emissions amplify the increase in OH under a 2 K idealized warming scenario through their contribution to OH production via the $\text{NO} + \text{HO}_2$ reaction. A 23% enhancement in SNO_x emissions driven by the +2 K surface warming leads to an increase in OH of 1.2%. This increase is driven by an enhancement of $1.1 \times 10^4 \text{ molec cm}^{-3} \text{ s}^{-1}$ in OH production, primarily attributed to the $\text{O}^1\text{D} + \text{H}_2\text{O}$ reaction and the $\text{NO} + \text{HO}_2$ reaction (Figure S7a in Supporting Information S1 and Figure 4a). Compared to the OH production pathways, the influence of temperature-sensitive SNO_x emissions on OH loss pathways is minimal (Figure S7b in Supporting Information S1).

In contrast to SNO_x emissions, incorporating temperature-sensitive BVOC emissions significantly offsets the enhancement in OH under idealized warming, primarily by increasing the OH loss frequency. Despite opposing effects on the OH response to the 2 K idealized warming, the influence of BVOC emissions is much greater than that of SNO_x emissions. A 37% increase in BVOC emissions leads to OH reductions of 5.2% by enhancing OH loss by 0.098 s^{-1} . OH reactions with BVOCs are the largest contributors (Figure 4b), driving a 39% increase in OH loss frequency. Reactions with CO and oxygenated VOCs, both products of local BVOC oxidation, contribute an additional 32% and 11% of the increase in OH loss frequency, respectively. Notably the increase in OH loss frequency due to reactions with NMVOCs is located in the regions with abundant biogenic sources (Figure 5f). However, the increase in OH loss frequency through the reaction with CO occurs over a wider region due to its longer chemical lifetime that allows CO to undergo long range transport (Figure S6b in Supporting Information S1). Temperature-sensitive biogenic emissions also influence OH production. The global average OH production rates increase by $2.0 \times 10^4 \text{ molec cm}^{-3} \text{ s}^{-1}$, driven by the enhanced $\text{NO} + \text{HO}_2$ reaction, $\text{O}_3 + \text{HO}_2$ reaction and H_2O_2 photolysis (Figure 4a).

3.3. Additivity

The impact of individual processes on the OH response to warming is largely additive, as the non-linear effects between different drivers are relatively small. Accounting for the impact of moistening, along with the temperature sensitivity of both SNO_x and BVOC emissions, results in a combined 2.5% increase in OH and a 5.5% increase in $L(\text{CH}_4)$ (Figures 4c and 4d). Summing up the independent contribution from the moistening, SNO_x

and BVOC emissions yields an OH increase of 2.6% and a L (CH₄) increase of 5.7% due to idealized 2 K warming, slightly higher than the Aqua-chem simulation with combined impacts.

4. Summary and Discussion

With the full complexity of atmospheric chemistry as represented in a fully coupled chemistry-climate model, we investigate the extent to which simplifying climate dynamics and emissions can reveal process-level insights regarding regional variations of OH chemistry. With CESM2 Aqua-chem, we extend the idealized aquaplanet configuration of CESM2-CAM6 to include atmospheric chemistry. In Aqua-chem, we prescribe idealized zonally-averaged and hemispherically symmetric SSTs, and simplify both the insolation and emissions to annual-mean distributions to conduct perpetual-equinox simulations. Compared to the reference CAM-chem model, the Aqua-chem model accelerates the simulation time by an order of magnitude while successfully reproducing the spatial distribution and the partitioning of OH budgets. Any discrepancies between the idealization and full complexity model provide insights into the importance of processes neglected in the simplified representation.

We then use Aqua-chem to analyze the response of the OH budget to an idealized 2 K surface warming. We examine the competing factors contributing to OH changes, including moistening and temperature-sensitive SNO_x and BVOC emissions. The most significant effect of increased surface temperatures on OH is the rise in primary OH production due to higher water vapor concentrations. In contrast, the OH sink due to BVOC emissions offsets the faster OH production rate driven by moistening and SNO_x emissions.

It is worth noting that we only account for the temperature sensitivity of SNO_x and BVOC emissions as described in the emission schemes widely used in global atmospheric chemistry models. However, the responses of natural emissions to a warmer climate remain highly uncertain due to their sensitivity to various environmental factors beyond temperature. For instance, future elevated CO₂ concentrations may reduce BVOC emission rates by closing stomata (Heald et al., 2009; Rosenstiel et al., 2003), offsetting the temperature-driven increase in BVOC emissions. Future changes in natural emissions also depend on climate mitigation strategies, such as reforestation and afforestation. Therefore, the simulated OH response to a 2 K increase in SST does not necessarily reflect future changes in OH under any specific global change scenario, but rather reveals the fundamental drivers of the selected processes examined here. In addition, changes in lightning NO emissions and wetland CH₄ emissions under climate change are uncertain, and would further impact the OH budget along with SNO_x and BVOC emissions. Our idealized approach underlines the need to accurately predict future changes in natural emissions to confidently estimate future OH and consequently CH₄ budget.

We also envision broader applications of Aqua-chem. Although its simplifications in climate dynamics and emissions indicate that Aqua-chem cannot fully capture all aspects of chemistry-climate interactions and therefore cannot replace a fully coupled chemistry-climate model, it provides a computationally efficient framework for developing process-level understanding. Therefore, Aqua-chem can be used to reveal the mechanisms driving complex interactions between climate and chemistry, such as the coupling between aerosols and cloud formation. As OH is the important determinant of secondary aerosol production, the aerosol-cloud interactions are linked to OH chemistry but are challenging and computationally expensive to disentangle in a full complexity model.

Conflict of Interest

The authors declare no conflicts of interest relevant to this study.

Availability Statement

The analysis data sets and codes are available at Zhu et al. (2025a). The Aqua-chem source code and inputs are available at Zhu et al. (2025b).

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